# CUSTOMER & SALES DATA ANALYSIS SYSTEM

## Internship Final Project Report

## 1. Title Page

- Project Title: Customer & Sales Data Analysis System
- Domain: Retail Sales Analytics, Customer Segmentation & Business Intelligence
- Submitted By: Kavyasree
- Role: Data Analytics Intern
- Tools Used: Python (Pandas, NumPy), Microsoft Power BI Desktop, DAX, Excel/CSV
- Dataset Scope: 9,800 retail transaction records across 793 unique customers
- Date: October 2026

## 2. Abstract

This project focuses on analysing retail sales and customer data. First, I used Python to clean a dataset of 9,800 records and group 793 customers into 4 categories based on how often and how much they buy. Next, I imported the clean data into Microsoft Power BI to build an interactive dashboard. The dashboard tracks important numbers such as Total Sales ($2.26M), Average Order Value ($459.48) and an Average Delivery Time of 3.96 days. Finally, the project provides simple business suggestions to keep loyal customers and help increase sales.

## 3. Introduction

In any retail business, thousands of sales transactions happen across different cities, products and customers. However, looking at large spreadsheets with raw data makes it difficult for business owners to know how the company is performing. To solve this, data analytics helps organize messy numbers into clear facts. In this project, I used Python to clean the sales data and calculated how fast items are delivered and how often customers shop. Then, I created an interactive Power BI dashboard so managers can easily see total sales, top products and customer shopping patterns on a single screen.

## 4. Problem Statement & Objectives

**Problem Statement:**

Retail companies collect thousands of sales records every day, but raw spreadsheets do not show which customer groups bring in the most money or which customers have stopped buying. Without clear visual reports, managers cannot make fast business decisions.

**Project Objectives:**

- Clean raw sales data using Python by fixing mixed dates and removing empty values.
- Calculate delivery days and group customers into 4 segments using RFM analysis.
- Build an interactive Power BI dashboard tracking Total Sales ($2.26M) and Top Products.
- Provide simple business suggestions to improve customer retention and revenue.

## 5. Dataset Description

- Source Data: Raw sales dataset (sales data.csv) transformed into the final cleaned dataset (cleaned_sales_rfm.csv).
- Data Size: 9,800 transaction records covering 793 unique customers.
- Main Columns Used:
- Order Details: Order ID, Order Date, Ship Date, Ship Mode
- Customer Details: Customer ID, Customer Name, Segment (Consumer, Corporate, Home Office)
- Location Details: Country, City, State, Region (West, East, Central, South)
- Product Details: Category, Sub-Category, Product Name
- Sales (Revenue per order item)
- New Columns Created:
- Shipping _ Days: Number of days taken to deliver the order (Ship Date – Order Date).
- Customer _ Segment: The 4 customer groups created using RFM scores (Champions, Recent Growth, At Risk, Hibernating).

## 6. Data Cleaning & Preprocessing

- Handled Missing Values: Checked the dataset and removed rows with missing Customer IDs, dates, or sales values so all records are complete and accurate.
- Standardized Dates: Converted both Order Date and Ship Date into a proper date format using Python, fixing any inconsistent date styles.
- Cleaned Numeric Sales: Removed currency symbols ($) and commas from sales figures and converted them into clean decimal numbers for calculation.
- Filtered Out Invalid Records: Kept only real purchase records where sales were greater than zero, removing zero or negative return entries.
- Calculated Delivery Time: Created the new column Shipping _ Days by subtracting Order Date from Ship Date to measure delivery speed.

## 7. Exploratory Data Analysis (EDA)

- Total Sales Volume: The dataset contains a total sales revenue of $2.26M ($2,261,536.78) across all orders.

- Total Orders & Customers: There are 4,922 unique orders placed by 793 unique customers.
- Customer Segments Breakdown: Customers belong to three main buyer types: Consumer (~51%), Corporate (~30%) and Home Office (~19%).
- Average Delivery Performance: Orders take an average of 3.96 days to be shipped from the order date.
- Sales Trend Pattern: Sales show a clear seasonal pattern, with transactions steadily rising and reaching the highest peaks towards the end of the year (September to December).

## 8. Statistical Analysis

- Total Sales Revenue: $2,261,536.78 ($2.26M) across all transactions.
- Average Order Value (AOV): $459.48 per distinct order (calculated by dividing Total Sales by 4,922 unique orders).
- Average Transaction Value: $230.77 per individual line item across all 9,800 rows.
- Average fulfillment Lead Time: 3.96 days from order placement to shipping date.
- Data Distribution (Skewness): The sales numbers have a right skew (positive skewness). This happens because most orders are lower-priced office supplies, but a few very large enterprise technology orders pull the average up.

## 9. Customer Analytics

- Customer Diversity: The business serves 793 unique customers across Consumer, Corporate and Home Office groups.
- High Customer Value: Repeat customers generate the majority of total revenue, showing strong loyalty.
- Purchase Frequency: Many customers place repeat orders across multiple years, proving steady demand.
- Inactive Customer Alert: Analysis shows several buyers have not made a purchase in over 6 months (180 + days), highlighting a need for re-engagement offers.

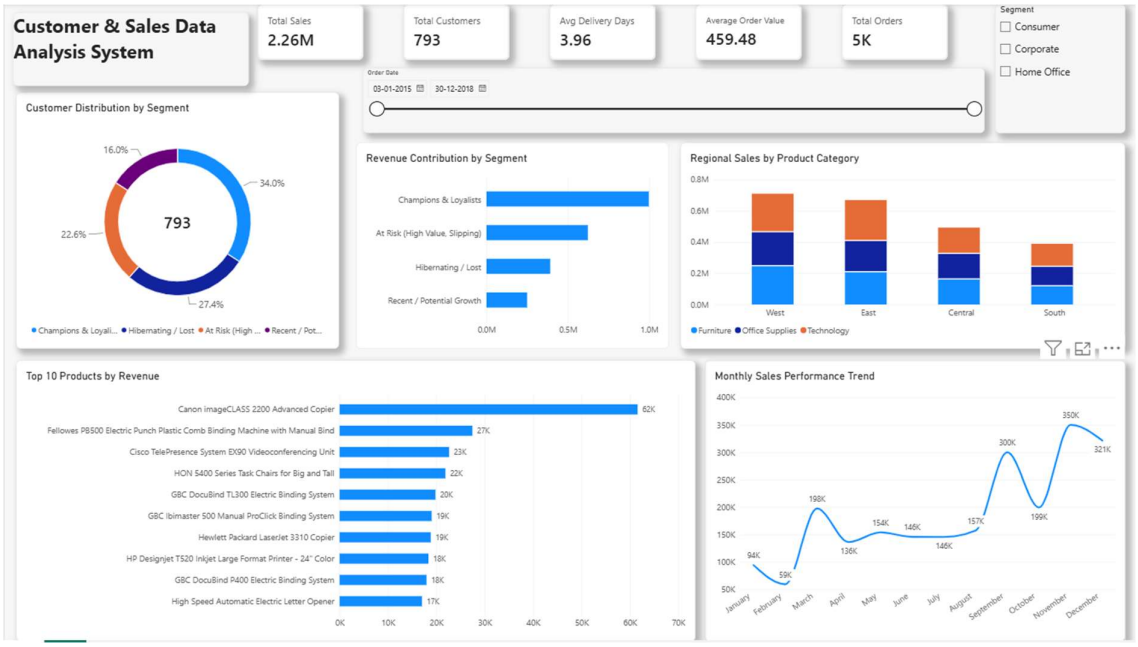
## 10. Sales and Product Analysis

- Regional Sales Leader: The West region generates the highest total revenue ($0.73M).
- Top Product Categories: Technology and Office Supplies contribute the highest revenue, while Furniture provides consistent, steady order volume.
- Top-Grossing Product: The highest revenue-generating product in the store is the Canon image class 2200 Advanced Copier (totalling over $61,000 in sales).
- Yearly Revenue Growth: Total sales increased steadily year-over-year, showing healthy business expansion across all customer accounts.

## 11. Customer Segmentation Method (RFM)

- RFM Metric Calculation: In Python, three metrics were calculated for each customer: Recency (days since last purchase), Frequency (total distinct orders placed) and Monetary (total spend in dollars).
- Champions & Loyalists: Customers with low recency (shopped recently), high order count and high total spend. They generate the biggest share of company revenue.
- Recent / Potential Growth: Customers who placed orders recently but with fewer overall orders. They have high potential to become loyal buyers.
- At Risk: High-value customers who spent significant money in the past but have not purchased recently. They require re-engagement offers.
- Hibernating / Lost: Customers who made very few purchases a long time ago and have stayed inactive.

## 12. Dashboard Screenshots and Explanation




- Executive KPI Cards (Top Banner): Display high-level business totals at a glance, including Total Sales ($2.26M), Unique Customers (793), Average Delivery Days (3.96 Days), Average Order Value ($459.48) and Total Orders (~5K).
- Customer Distribution by Segment (Donut Chart): Shows the proportion of 793 customers categorized into four RFM groups (Champions & Loyalists, At Risk, Hibernating and Recent / Potential Growth).
- Revenue Contribution by Segment (Bar Chart): Clearly proves that Champions and Loyalists generate the largest portion of total revenue.

- Regional Sales by Product Category (Stacked Column Chart): Tracks sales performance across West, East, Central and South regions broken down by Furniture, Office Supplies and Technology.
- Top 10 Products by Revenue (Horizontal Bar Chart): Ranks the top revenue-driving products, led by enterprise hardware such as copiers and binding systems.
- Monthly Sales Performance Trend (Line Chart): Illustrates seasonal demand variations and steady sales acceleration toward year-end peaks.
- Interactive Slicers: Date range slider and segment filters enable users to interactively filter and drill down into specific timeframes or customer groups.

## 13. Results and Findings

- High Revenue from Loyal Buyers: The Champions & Loyalists customer group makes up about 34% of customers but brings in over 46% (more than $1.05M) of total company revenue.
- Valuable Customers at Risk: There are 179 high-value buyers in the At Risk category who previously spent over $580,000, meaning the business must re-engage them before they stop buying completely.
- Consistent Delivery Time: The average fulfilment lead time across all orders is 3.96 days, showing reliable shipping operations.
- Leading Sales Regions: The West and East territories are the strongest markets, together generating over 60% of total sales revenue.

## 14. Insights and Recommendations

- Reward Loyal Champions: Since Champions & Loyalists generate over 46% of total revenue, offer them VIP perks and early access to new products so they keep buying.
- Win-Back Campaigns for At-Risk Customers: Send special re-engagement discount emails to the 179 high-value At Risk customers before they stop shopping completely.
- Product Bundling to Grow Basket Size: Combine high-selling items (like office supplies) with enterprise equipment to increase the Average Order Value above $459.48.
- Focus on Weaker Sales Regions: The South region has the lowest sales ($0.40M), so increasing targeted promotions in southern cities will help boost growth.

## 15. Limitations and Future Scope

- Lack of Profit Data: The dataset records gross sales revenue but does not include product manufacturing costs or net profit margins.

- No Return Tracking: There is no specific column tracking customer product returns or refunds.
- Predictive Machine Learning: In the future, machine learning models (like Churn Prediction or K-Means clustering) can be added to predict customer behaviour earlier.
- Sales Forecasting: Time-series forecasting models can be built to help store managers predict inventory demand for the next quarter.

## 16. Conclusion and References

- Conclusion: This project successfully built an end-to-end data analysis system. By using Python for data cleaning and RFM segmentation and Power BI for interactive dashboard visuals, business managers can easily track key metrics ($2.26M sales, 793 customers) and make smart decisions to retain high-value buyers.
- References:

    1. Superstore Retail Sales Dataset (sales data.csv)
    2. Microsoft Power BI Desktop Documentation & DAX Reference Guide
    3. Python Pandas Library Documentation (pandas.pydata.org)
    4. Customer & Sales Data Analysis System Internship Project Guide